Fiscal Year:	FY 2010	**Task Last Updated:**	FY 05/12/2010
PI Name:	Songer, Jocelyn Ph.D.		
Project Title:	Investigating High-Frequency Vestibular Function: A Potential Flight Diagnostic		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Sensorimotor Adaptation Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	jocelyns@paradoxical.net	**Fax:**	FY
PI Organization Type:	PUBLIC SERVICE	**Phone:**	617-573-3176
Organization Name:	Massachusetts Eye and Ear Infirmary		
PI Address 1:	Eaton-Peabody Laboratory		
PI Address 2:	243 Charles Street		
PI Web Page:			
City:	Boston	**State:**	MA
Zip Code:	02114-3002	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2007 NSBRI-RFA-07-02 Postdoctoral Fellowships
Start Date:	12/01/2007	**End Date:**	11/30/2009
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Merfeld, Daniel (MENTOR/Massachusetts Eye and Ear Infirmary) Eatock, Ruth (MENTOR/Massachusetts Eye and Ear Infirmary)		
Grant/Contract No.:	NCC 9-58-PF01402		
Performance Goal No.:			
Performance Goal Text:			
	POSTDOCTORAL FELLOWSHIP The otolith organs, the utricle and saccule, play a role in a number of clinical vestibular disorders and have been hypothesized to play a role in some of the vestibular dysfunction experienced by astronauts during spaceflight and after re-entry. The mammalian saccule, which senses gravity, accelerations, and loud acoustic stimuli, is one of the least studied organs of the inner ear. Although there are morphological changes in the saccular epithelium (to both type I and type II hair cells) as a result of spaceflight, little is understood about how these changes might affect saccular physiology. Improving our understanding of the saccule and its frequency characteristics is of particular interest because measurements of saccular sensitivity in response to loud sounds are starting to be used in the clinic. The goal of this project is to improve our understanding of the mammalian saccule by focusing on the frequency characteristics of		

Task Description:	the saccular epithelium. The original project had two aims; the first aim was to characterize the frequency tuning of saccular reflexes in response to tones using both vestibular evoked myogenic potentials, and vestibulo-ocular reflexes in adult rats, and the second aim was to evaluate the frequency characteristics of saccular hair cells by receptor potentials and transduction currents from the central (striolar) and peripheral (extrastriolar) zones of the saccular epithelium. Based on preliminary results from both projects, we decided to focus our efforts on aim 2. We hypothesize that within the saccular epithelium hair cell frequency sensitivity will vary with hair cell type and zone, with some cells sensitive to higher frequency stimuli, including acoustic frequencies and others sensitive to lower frequencies, including frequencies of voluntary head motions. To test this hypothesis we compare voltage-gated currents, receptor potentials and transduction currents from type I and type II hair cells from both striolar and extrastriolar zones of the saccular epithelium using whole-cell patch recording methods in the semi-intact rat saccule. Preliminary results comparing striolar type I hair cells and extrastriolar type II hair cells suggest differences. Extrastriolar type II hair cells have a midpoint voltage for sodium channel inactivation (-74±2mV, n=6) that is significantly more positive for type II extrastriolar hair cells than striolar type I hair cells (-93±1mV, n=19). Additionally, striolar type I hair cells have a faster fast time constant of adaptation (7.8±2ms, n=7) in response to step bundle deflections when fit with a double exponential than the extrastriolar type II hair cells (15±3ms, n=5), a smaller operating range (1.1±0.2µm, n=8 compared to 1.7±0.1µm, n=5) and a higher upper cutoff frequency (60±10Hz, n=9 compared to 29±3Hz, n=4) when stimulated with sinusoidal bundle deflections from 2Hz to 100Hz and fit with a double exponential. These preliminary data are consistent with our hypothesis that frequency filtering may differ between hair cell types and zones, especially when looking at the fast component of adaptation. Evaluating responses to higher frequencies (up to 500 Hz) and additional cell types and zones we will be able to more fully characterize these differences. Characterization of the frequency tuning of hair cells in the saccular epithelium will give us insight into what drives vestibular afferent firing patterns, improve our understanding of peripheral contributions to vestibular reflexes, may improve and expand our understanding of the frequency characteristics of VEMPs, and may provide insight into how morphological changes in the saccular epithelium resulting from spaceflight contribute to the vestibular dysfunction experienced by astronauts.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Understanding the frequency tuning of the mammalian saccule will help us answer basic science questions pertaining to the specialization of the saccule and may allow us to improve existing clinical measures of saccular function such as measures of vestibular evoked myogenic potentials.
Task Progress:	The project, as defined two years ago, was broad in scope, with two very different aims drawing on the training possibilities present in each of the co-mentors' laboratories (Drs. Merfeld and Eatock). The first aim was to characterize, in adult rats, the frequency tuning of saccular reflexes in response to acoustic tones by measuring vestibulo-ocular reflexes and VEMPs. The second aim was to evaluate the frequency characteristics of saccular hair cells by recording receptor potentials and transduction currents from individual hair cells in an intact saccular epithelium from both the striolar and extrastriolar zones. Our ultimate goal is to compare the frequency tuning and organization of the saccular epithelium to that of vestibular reflexes, gaining insight into saccular function that could lead to the improvement of existing diagnostics and the development of novel diagnostics. Two years ago, each project was at an early stage of development, such that significant start-up time was required. However, my involvement in both projects from such an early stage provided invaluable training and experience in diverse skills, ranging from eye-coil surgeries in squirrel monkeys to the micro-dissection of postnatal rat saccules, and from the development of a unique sound system for delivery of sinusoidal tone bursts (50 Hz – 4 kHz, 60-125 dB SPL) to awake squirrel monkeys to the development of micromechanical stimulators to deflect saccular hair bundles with sinusoidal stimuli (2 Hz – 500 Hz, 0.5 – 2.0 µm) and step stimuli. Much of the first year of the project was devoted to setting up and debugging these two experiments. By year two, data collection and analysis had begun in earnest on both projects. Our preliminary sound-evoked VOR data from the squirrel monkey suggested that variations in eye position at the onset of the tone, and other variations in the baseline noise, would obscure any sound-evoked potentials, leading us to conclude that experiments might better be conducted on subjects that had larger eye movements and which could be trained to fixate on a target during measurements. On the other hand, the project on saccular hair cells was starting to work and we were recording large transduction currents in response to our sinusoidal stimuli. We settled on evaluating contributions of saccular hair cell properties to acoustic and vestibular tuning by comparing frequency dependence of transduction currents and receptor potentials across hair cell types and locations, and by investigating the sources of underlying tuning in properties of transduction and voltage-gated ion channels. This narrowing of focus accelerated progress. My preliminary data now suggest that frequency filtering by hair cells varies as a function of hair cell type and zone in a manner consistent with our underlying hypothesis. These data are being presented in a talk and a poster at the 2010 meeting of the Association for Research in Otolaryngology.
Bibliography Type:	Description: (Last Updated: 09/06/2019)
Abstracts for Journals and Proceedings	Eatock RA, Kalluri R, Songer JE. "The organization of signals in mammalian otolith organs." 33rd MidWinter Meeting, Association for Research in Otolaryngology, Anaheim, California, February 6-10, 2010. 33rd Annual Midwinter Research Meeting, Abstract Book, February 2010. , Feb-2010
Abstracts for Journals and Proceedings	Songer JE, Eatock RA. "Frequency characteristics of the mammalian saccular epithelium." 2010 NASA Human Research Program Investigators' Workshop, Houston, Texas, February 3 - 5, 2010. 2010 NASA Human Research Program Investigators' Workshop Program, February 2010. , Feb-2010
Abstracts for Journals and Proceedings	Songer JE, Eatock RA. "Transduction in the mammalian saccule." 33rd MidWinter Meeting, Association for Research in Otolaryngology, Anaheim, California, February 6-10, 2010. 33rd Annual Midwinter Research Meeting, Association for Research in Otolaryngology, February 2010. , Feb-2010

Articles in Peer-reviewed Journals	Eatock RA, Songer JE. "Vestibular hair cells and afferents: Two channels for head motion signals." Annu Rev Neurosci. 2011;34:501-34. https://doi.org/10.1146/annurev-neuro-061010-113710 ; PubMed PMID: 21469959 , Jul-2011